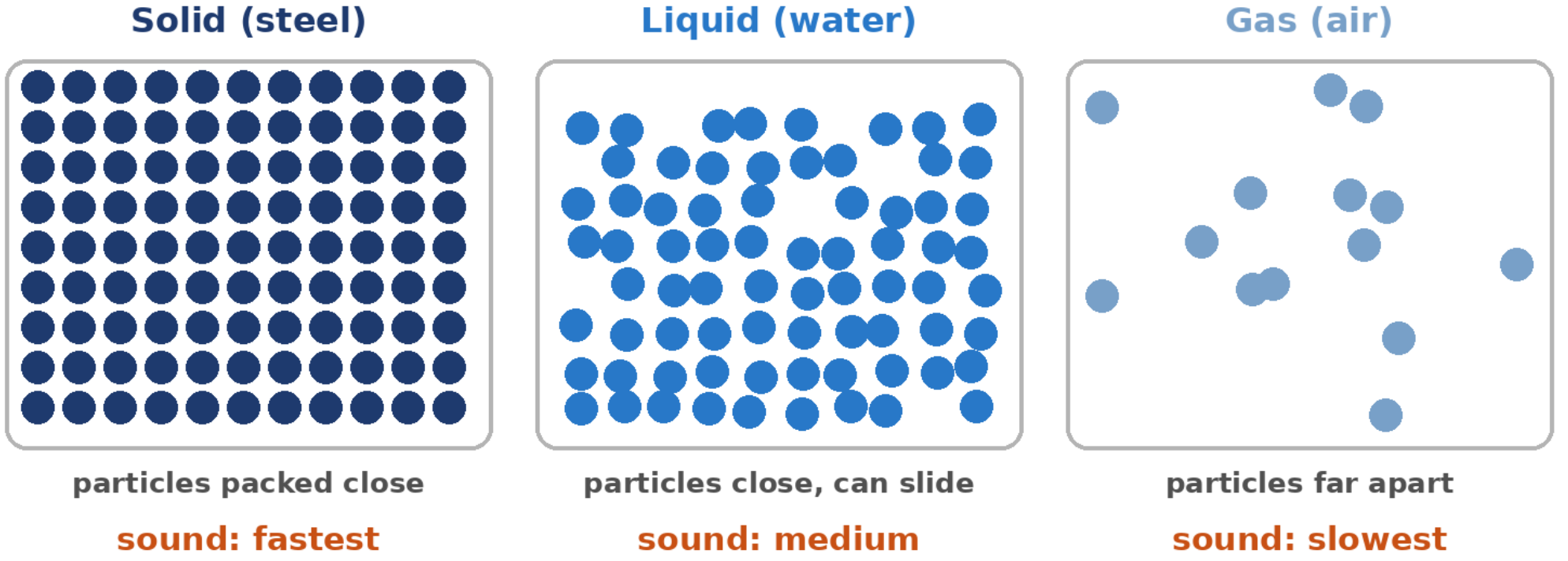
# 4.12 Explore Sound in a Medium

What changes the speed of sound? What does not?



Think About It

# Your friend taps a metal railing at the far end of a bridge. Which tap reaches you first?

Single choice poll

**A.** The sound through the air

Single choice poll

**B.** The sound through the railing

Single choice poll

**C.** Both at the same time

Single choice poll

**D.** Neither: sound can't travel in metal

# Today's Goals

I can explain how loudness and pitch show up in a sound wave.

I can explain why sound is fastest in solids.

I can tell what changes wave speed and what does not.

I can explain why there is no sound in space.

**Rate each goal:**

1 Not yet

2 Getting there

3 I can teach it

# CASE #3 • WEDNESDAY

## Is the lightning more than 1 km away?

**WAVE WORD:** *Speed of sound*

How fast sound travels: about 340 m/s in air.

### MYSTERY

You see a lightning flash, then hear thunder 3 seconds later. Light reaches you almost instantly, but sound travels through the air at about 340 meters per second. Distance = speed × time. Is the lightning more than 1 km away?

### CLUES

- Speed: 340 m/s
- Time: 3 s
- 1 km = 1,000 m







### CLAIM

Is the lightning more than 1 km away?

☐ YES ☐ NO

### EVIDENCE

Solve:

**340 × 3 = _____**

The lightning is _____ m away.

### REASONING

Finish the sentence:

The signs are _____, so the answer is _____. 1,020 m is _____ than 1,000 m (1 km).

# Today's Plan

## Part 1: OpenSciEd Sound Sim

Loud vs. soft, high vs. low. Watch the particles.

**Q1–Q3 · about 15 min**

## Part 2: Wave Speed Lab

7 missions + Fact or Myth. Earn a code and certificate.

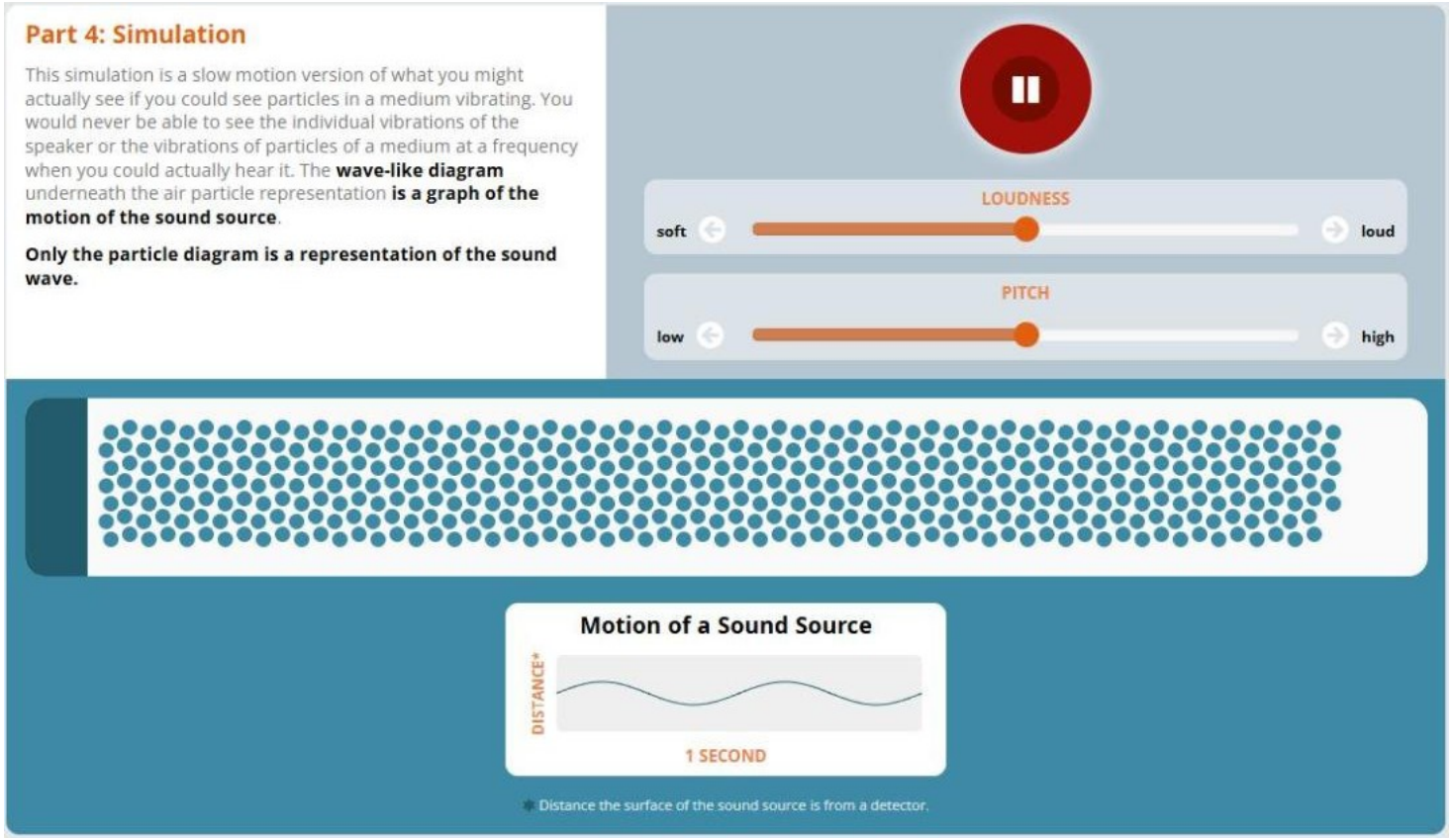
**Q4–Q7 · rest of class**

## Not done?

The Wave Speed Lab saves your progress in this browser. Finish it on your own.

**Keep the same computer**

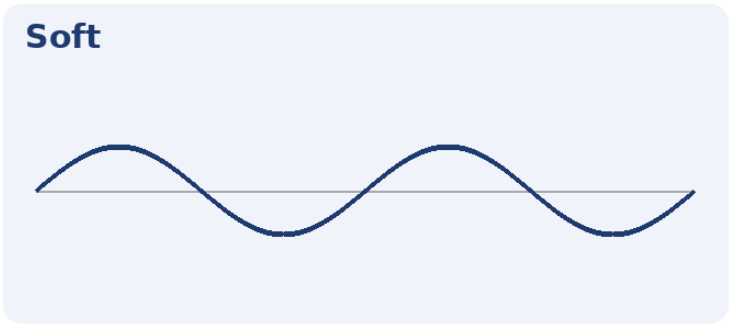
# Part 1: How Does Sound Move Through a Medium?

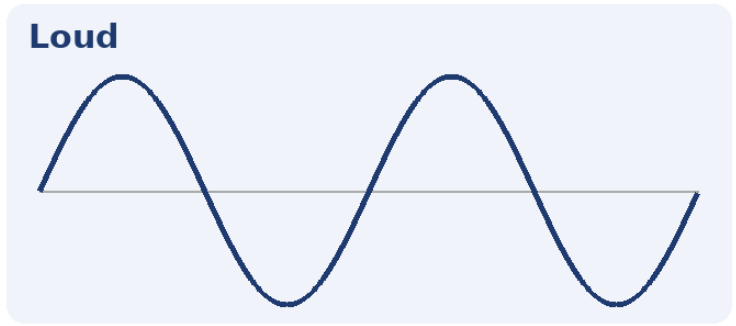


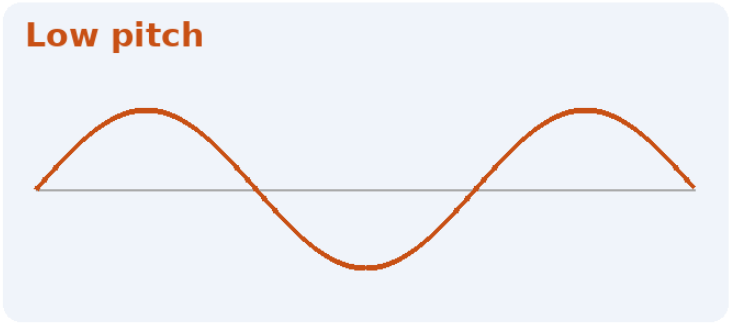

▶️ Open the OpenSciEd sound sim

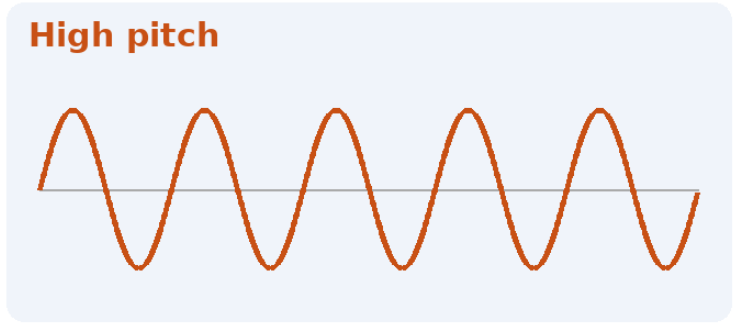
1. Press Play.
2. Move ONLY the Loudness slider. Watch the graph and the particles.
3. Reset. Move ONLY the Pitch slider.
4. Do the particles travel to the end, or vibrate back and forth?

# Reading Sound Graphs



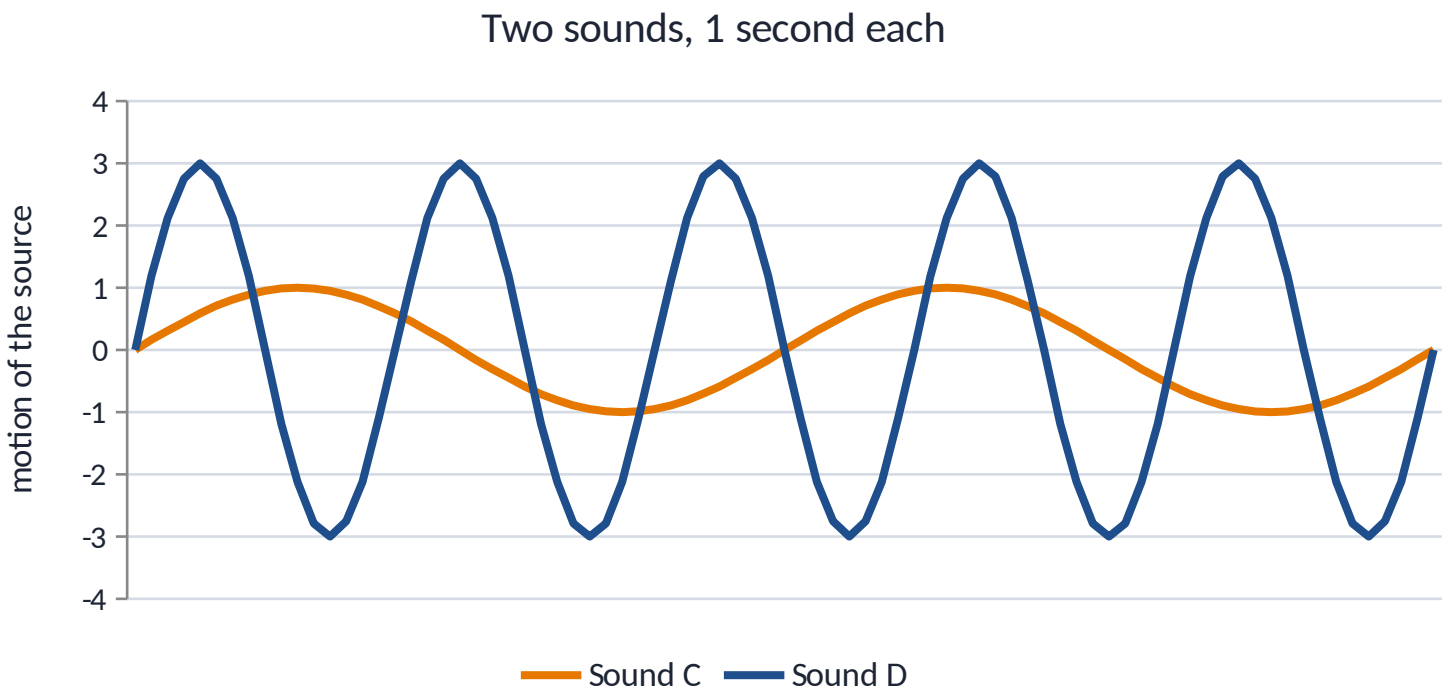

**Taller wave** = bigger amplitude = louder = more energy.

**More waves in 1 second** = higher frequency = higher pitch.

Neither one changes the speed. Only the medium does.

**Read the HEIGHT for loudness. Count the WAVES for pitch.**

## Graph Poll




Both graphs show 1 second.

**Which sound is LOUDER and has the HIGHER pitch?**

Single choice poll

**A.** Sound C

Single choice poll

**B.** Sound D

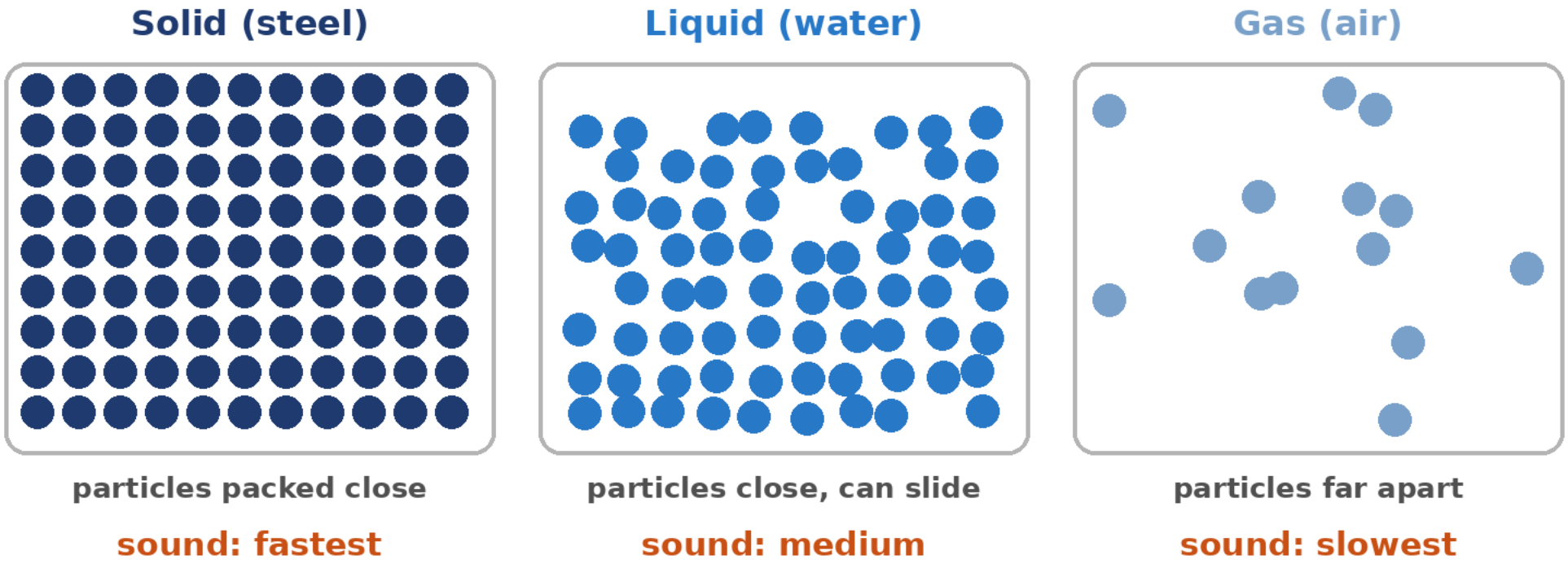
Single choice poll

**C.** They are the same

Single choice poll

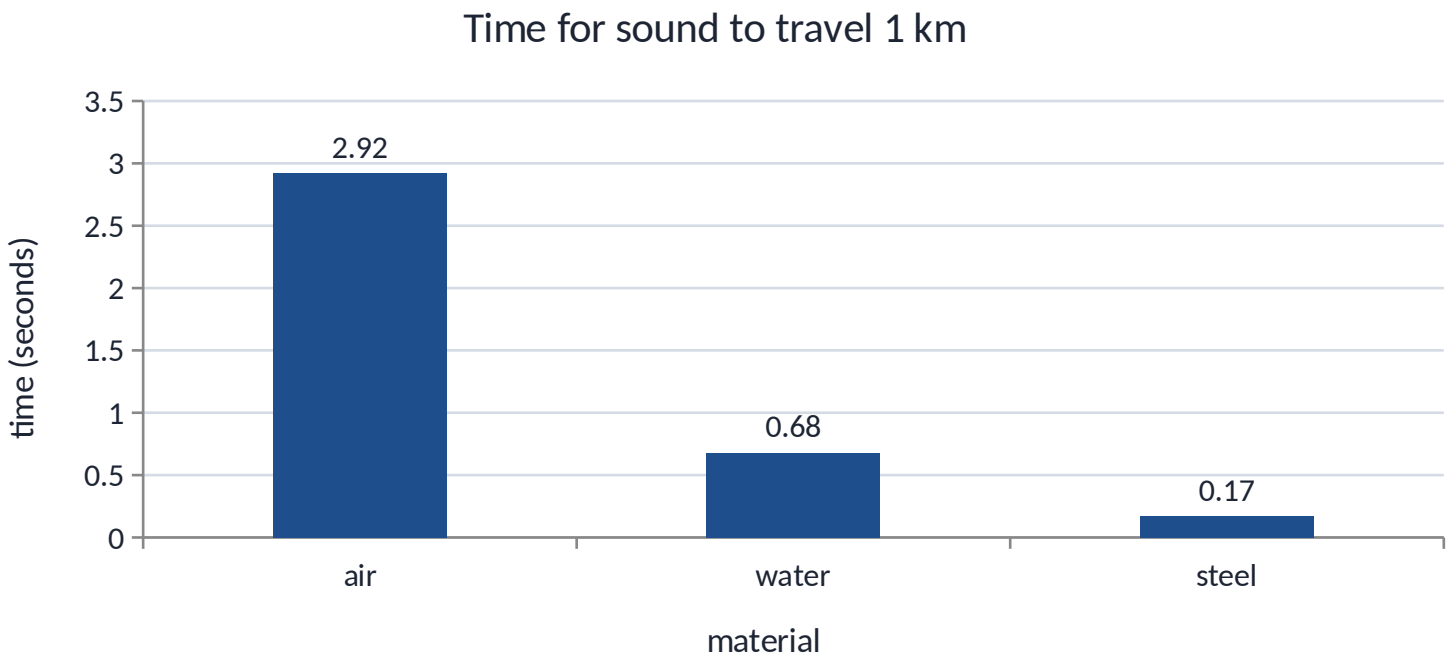
**D.** You can't tell

# Sound and the Medium




**Particles packed close together bump the next ones quickly, so sound is fastest in solids. Space has no particles, so there is no medium and no sound.**

## Graph Poll




Read the graph.

**You make a sound 1 km away. Through which material does it reach you FIRST?**

Single choice poll

**A.** air

Single choice poll

**B.** water

Single choice poll

**C.** steel

Single choice poll

**D.** all at the same time

**Who Is Right?**

## Maya says, "If I yell LOUDER, my voice will reach my friend across the field faster." Who is right?

**Ava says:**

*"Maya is right. Louder sounds are faster."*

**Ben says:**

*"Maya is wrong. Louder means bigger amplitude, but only the medium changes the speed."*

**Cam says:**

*"Maya is right. More energy always means more speed."*

**Dee says:**

*"Maya is wrong. Quiet sounds are faster."*

Single choice poll

**A.** Ava

Single choice poll

**B.** Ben

Single choice poll

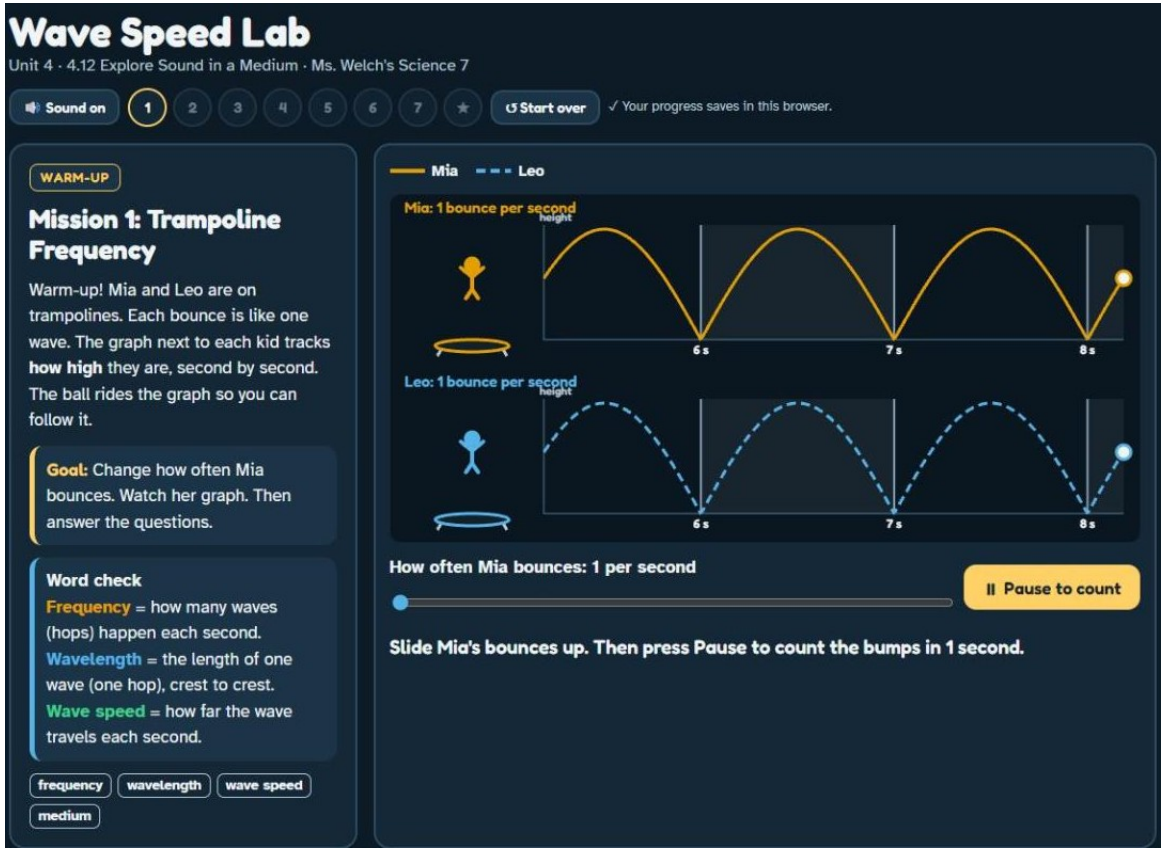
**C.** Cam

Single choice poll

**D.** Dee

Classkick: 4.12 Part 2 · Q4–Q6

# Part 2: Wave Speed Lab




▶️ Open the Wave Speed Lab

## Use the rest of class

1. Trampoline Frequency
2. Who Will Win? (predict)
3. The Big Race
4. Hops Make Waves
5. Rope Race (4 rounds)
6. Same Note, New Medium
7. Tin Can Phones

★ Fact or Myth?

# Finish: Code and Certificate

## 1. Type your code

Finish every mission. Type the completion code in the Q7 box.

## 2. Copy the certificate

Click Copy certificate in the lab (or snip it: Windows + Shift + S).

## 3. Paste it

Click the box on the Classkick slide and press Ctrl + V.

# Key Ideas

1. Sound is a longitudinal, mechanical wave. It needs a medium.
2. Loudness = amplitude (more energy). Pitch = frequency.
3. Only the medium changes wave speed. Frequency, wavelength, and amplitude do not.
4. Sound is fastest in solids, slower in liquids, slowest in gases.

**Re-rate your goals. Not done? Finish the Wave Speed Lab and paste your certificate.**